Dynamic Recovery Window Model in Pediatric Surgical Emergencies Using Lactate Clearance, Capillary Refill, and Delay Sensitive Perfusion State Transitions

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Abstract

Children requiring emergency surgery may deteriorate while awaiting definitive intervention, but the consequences of delay depend on evolving perfusion and recovery capacity rather than time alone. We developed a literature-parameterized Dynamic Recovery Window model to examine when a simulated pediatric perfusion trajectory remains recoverable as intervention is delayed. This mathematical proof-of-concept model integrates lactate clearance, capillary refill, perfusion reserve, recovery reserve, injury accumulation, and delay load within a nonlinear dynamical system. Continuous physiological states inform a four-regime continuous-time Markov process representing compensated, vulnerable, decompensating, and refractory states, while a finite-horizon optimal-stopping layer formalizes the trade-off between immediate intervention and continued waiting. Model behavior was evaluated through numerical solver verification, delay continuation, scenario perturbation, global sensitivity analyses, and synthetic model-recovery testing. With intervention at 1 hour, lactate decreased from 4.20 to 1.52 mmol/L and capillary refill from 3.10 to 1.52 seconds over 10 hours, while perfusion reserve increased from 0.580 to 0.998 and recovery reserve from 0.620 to 0.915. Under the reference assumptions, trajectories remained reversible through a 4.5-hour intervention delay, became partially reversible at 5.0-5.5 hours, and entered the irreversible model class at 6.0 hours. These boundaries are simulation outputs rather than clinical thresholds. Across 1,000 Latin-hypercube parameter sets evaluated at a 2-hour delay, the terminal recovery index had a median of 0.932 (2.5^{th}-97.5^{th} percentile, 0.840-0.975). Recovery-reserve restoration (PRCC 0.933) and post-intervention injury clearance (PRCC 0.923) were the strongest favorable determinants, whereas reserve injury (PRCC -0.833) and injury generation (PRCC -0.775) were the strongest adverse determinants. Synthetic model-recovery analysis supported computational recovery of the generating delay-sensitive structure but was not interpreted as clinical validation. The dynamic recovery window provides a mechanistic and mathematically testable representation of how intervention delay may interact with evolving perfusion, injury, and recovery capacity. Its current purpose is to guide variable selection, repeated-measure study design, parameter calibration, and future clinical validation rather than bedside decision-making. Patient-level calibration and external validation are required before any modeled recovery boundary can be interpreted as a treatment threshold.

1. Introduction

Emergency pediatric surgery presents a time-sensitive clinical problem in which physiological deterioration may continue while a child awaits transport, diagnostic evaluation, operating-room access, anesthesia, or definitive source control. This challenge is particularly important because children requiring emergency surgery may present with substantial heterogeneity in baseline physiological condition, disease severity, and access to timely surgical care (GlobalSurg Collaborative, 2016, Firomsa et al., 2018 and PaedSurg Africa Research Collaboration, 2021). Consequently, chronological delay alone may provide an incomplete representation of risk: the effect of waiting is likely to depend on the physiological state at presentation and the rate at which perfusion and recovery capacity subsequently deteriorate.

Clinical evidence supports the importance of operative timing without establishing a universal safe-delay threshold. In a risk-adjusted analysis of pediatric emergency surgery, Butterworth et al. (2023) reported that 47.9% of class-1 emergency operations were delayed beyond 60 minutes and that such delay remained associated with major morbidity or mortality after adjustment for clinical risk. Similarly, a prospective study of pediatric emergency surgery found that operative timing and duration contained prognostic information for postoperative mortality (Kaushal-Deep et al., 2019). These observations suggest that delay is clinically consequential, but they do not determine how an individual physiological trajectory evolves during the waiting period. A mathematical framework that represents delay as a dynamic process rather than a binary or static covariate may therefore provide complementary mechanistic insight.

Lactate and capillary refill time were selected as observable components of this framework because they provide complementary information about systemic and peripheral perfusion. Lactate has been repeatedly investigated as a marker of illness severity and resuscitation response in pediatric shock, while serial lactate measurements and lactate clearance may provide greater prognostic information than an isolated baseline measurement (Choudhary et al., 2017, Nazir et al., 2019 and Yuniar et al., 2023). Hemodynamic responses in pediatric shock are nevertheless heterogeneous, and normalization of one circulatory marker does not necessarily imply restoration of the broader physiological state (Yuliarto et al., 2022 and Obonyo et al., 2022). Capillary refill time provides a readily obtainable bedside measure of peripheral perfusion, prolonged capillary refill has also been associated with alterations in microcirculatory and endothelial glycocalyx markers in children with sepsis (Fernandez-Sarmiento et al., 2024). Accordingly, lactate and capillary refill should not be treated as interchangeable or individually deterministic measures of surgical timing.

The central methodological gap is therefore not simply the prediction of adverse outcome from individual biomarkers, but the representation of how observable perfusion markers, latent recovery capacity, accumulated injury, and intervention delay evolve jointly over time. In the present study, lactate and capillary refill are coupled to latent perfusion reserve, recovery reserve, and injury burden within a nonlinear dynamical system. These continuous states are further connected to latent physiological regimes to represent transitions from compensated toward increasingly compromised states. This structure is intended to distinguish an evolving physiological trajectory from a conventional static risk score and to provide a mechanistic basis for studying loss of recoverability.

The decision layer draws conceptually on financial engineering and sequential decision science. Recovery capacity is represented as a state that can be depleted or restored, physiological deterioration can transition between latent regimes, continued waiting carries an accumulating cost, and intervention

constitutes an irreversible timing decision. The analogy does not imply equivalence between biological and financial systems; rather, concepts such as reserve depletion, regime switching, uncertainty propagation, and optimal stopping provide a mathematical language for examining when continued waiting ceases to preserve recovery potential. Within this model, the dynamic recovery window is defined as the latest modeled intervention delay for which the simulated trajectory remains within a pre-specified recovery basin under a stated parameter set.

This study therefore develops and computationally evaluates a literature-parameterized Dynamic Recovery Window model for pediatric surgical emergencies. The primary objective is to determine whether lactate dynamics, capillary refill, latent perfusion and recovery reserves, injury accumulation, and intervention delay can be integrated into a mathematically coherent framework that generates a state-dependent boundary of recoverability. The model is presented as a theoretical proof-of-concept rather than a patient-level prediction or clinical decision tool. Its purpose is to generate testable hypotheses, identify parameters and repeated measurements that require clinical calibration, and establish a transparent framework for subsequent retrospective or prospective validation.

2. Methods

2.1 Study Design and Evidence Status

This study developed a theoretical mathematical-biology, computational-medicine, and decision-science framework for investigating time-dependent recovery potential in pediatric surgical emergencies. No participants were recruited, and no patient-level dataset was downloaded, fitted, or analysed. Published open-access studies of pediatric emergency surgery, shock, lactate kinetics, capillary refill, and hemodynamic response were used to select observable variables, establish directional physiological relationships, and define plausible reference conditions. Evidence supporting the clinical relevance of operative delay was drawn from pediatric emergency-surgery studies (Butterworth et al., 2023 and Kaushal-Deep et al., 2019), whereas the representation of lactate and peripheral perfusion was informed primarily by pediatric shock and sepsis studies (Choudhary et al., 2017, Fernandez-Sarmiento et al., and Obonyo et al., 2022). Published evidence informed the direction and plausible ranges of the model parameters. Where direct empirical estimates were unavailable, provisional values were specified to preserve the intended physiological relationships and were subsequently examined through sensitivity, uncertainty, and scenario analyses. These parameters therefore define the reference mathematical system and its perturbation space rather than patient-level effect estimates.

2.2 Clinical State Variables

The model state vector was defined as $x(t) = [L(t), C(t), P(t), R(t), I(t)]^{\mathrm{T}}$, with an additional externally defined delay-load process $D(t)$. Lactate and capillary refill time were treated as observable states, whereas perfusion reserve, recovery reserve, and injury burden were latent normalized states. Lactate was selected because serial lactate and lactate clearance have showed prognostic relevance in pediatric shock (Choudhary et al., 2017 and Nazir et al., 2019). Capillary refill was included as a complementary bedside marker of peripheral perfusion, supported by evidence

linking prolonged refill with microcirculatory abnormalities in pediatric sepsis (Fernandez-Sarmiento et al., 2024). These markers were not assumed to be interchangeable or sufficient individually to determine recovery.

Table 1. Continuous State Variables

Symbol	State	Scale	Interpretation
$L(t)$	Blood lactate	mmol/L	Observable metabolic/perfusion stress, initialized at 4.2 mmol/L in reference scenario.
$C(t)$	Capillary refill time	seconds	Observable peripheral perfusion marker, initialized at 3.1 s.
$P(t)$	Perfusion reserve	0-1	Latent ability to sustain effective tissue perfusion.
$R(t)$	Recovery reserve	0-1	Latent capacity to return toward a stable physiological state after intervention.
$I(t)$	Injury burden	0-1	Accumulated ischemic/metabolic injury.
$D(t)$	Delay load	0+	Time-dependent burden accrued before definitive intervention and decaying after intervention.

The reference lactate and capillary-refill values were modeling initial conditions rather than diagnostic thresholds. The latent states P(t), R(t), and I(t) do not currently correspond to directly measured clinical variables and would require explicit observation models in future patient-level calibration.

2.3 Hybrid Dynamic Model

The model comprises six interacting mechanisms rather than treating surgical delay as a single regression covariate. Lactate follows a production-clearance balance, with clearance increasing after intervention. Capillary refill follows a relaxation process determined by perfusion deficit and injury. Perfusion reserve is restored through recovery capacity and intervention but depleted by lactate burden, prolonged refill, and accumulated delay. Recovery reserve is restored under adequate perfusion and depleted by injury and delay. Injury accumulates under impaired perfusion and metabolic burden and clears more rapidly after intervention. Parameter uncertainty is common in dynamical systems biology models, particularly when individual kinetic parameters cannot be directly estimated from observed data (Gutenkunst et al., 2007). Finally, delay is represented as an exogenous time-varying load that increases before intervention and decays after intervention. The system was

$$\frac{dL}{dt} = a_L(1-P)(1+0.35I) - c_L(u)(L-1.5)$$

$$\frac{dC}{dt} = a_C[(1-P)+0.35I] - c_C(C-1.5) - 0.30u(C-1.5)$$

$$\frac{dP}{dt} = a_P\,R(1-P) - b_L\,L_n\,P - b_C\,C_n\,P - b_D\,D\,P + g_u\,u(1-P)$$

$$\frac{dR}{dt} = a_R\,P(1-R) - d_I\,I\,R - d_D\,D\,R + 0.10uP(1-R)$$

$$\frac{dI}{dt} = a_I[(1-P) + 0.30L_n + 0.25C_n](1-I) - c_I(u)RI$$

$D(t) = \min\left(\frac{t}{s_D}, 1.5\right)$ before intervention, $D(t) = \left(\frac{\tau}{s_D}\right)\exp[-0.7(t-\tau)]$ after intervention. The intervention indicator was defined as $u(t) = 0$ for $t < \tau$ and $u(t) = 1$ for $t \geq \tau$, where τ denotes the modeled intervention time. L_n and C_n denote normalized non-negative lactate and capillary-refill burdens used to place observable markers on scales compatible with the latent-state equations. Lactate and capillary-refill burdens were transformed into bounded normalized quantities before entering the perfusion, injury, and deterioration-propensity equations. Specifically,

$$L_n(t) = \text{clip}\left(\frac{L(t)-1.5}{4.5}, 0, 1.5\right),$$
$$C_n(t) = \text{clip}\left(\frac{C(t)-1.5}{3.5}, 0, 1.5\right),$$

Where $\text{clip}(x, a, b) = \min\{\max(x, a), b\}$. The lower reference value of 1.5 corresponds to the zero burden point used computationally, while the upper clipping bound limits extreme values from exerting unbounded influence on the latent-state equations. These transformations are model-scaling assumptions rather than clinically validated thresholds. The reference parameterization was specified to preserve the intended directional relationships among observable perfusion markers, latent reserve, injury accumulation, and intervention response. Twenty parameters were assigned predefined uncertainty ranges for global sensitivity analysis. These values were not estimated from patient-level data and should therefore be interpreted as provisional normalized coefficients defining the reference mathematical system and its perturbation space (Table 2).

Table 2. Reference model parameters and uncertainty ranges

Parameter	Reference	LHS range	Model interpretation
lactate_prod	0.48	0.30-0.68	Lactate production
lactate_clear	0.34	0.20-0.48	Pre-intervention lactate clearance
lactate_clear_post	0.62	0.45-0.82	Post-intervention lactate clearance
crt_impair	0.72	0.45-0.95	CRT impairment
crt_recover	0.58	0.38-0.80	CRT recovery
perfusion_restore	0.42	0.28-0.60	Baseline perfusion restoration
perfusion_lactate	0.20	0.10-0.32	Lactate-associated perfusion penalty
perfusion_crt	0.16	0.08-0.26	CRT-associated perfusion penalty
perfusion_delay	0.22	0.10-0.35	Delay-associated perfusion penalty
reperfusion_gain	0.95	0.65-1.25	Post-intervention reperfusion gain
reserve_recover	0.28	0.16-0.42	Recovery-reserve restoration
reserve_injury	0.34	0.20-0.50	Injury-associated reserve depletion
reserve_delay	0.18	0.08-0.30	Delay-associated reserve depletion
injury_gain	0.34	0.20-0.50	Injury generation

injury_clear	0.24	0.12-0.36	Baseline injury clearance
injury_clear_post	0.52	0.3-0.72	Post-intervention injury clearance
hazard_lactate	0.55	0.30-0.80	Lactate contribution to deterioration propensity
hazard_crt	0.48	0.25-0.72	CRT contribution to deterioration propensity
hazard_reserve	0.72	0.45-1.00	Low-reserve contribution to deterioration propensity
hazard_delay	0.62	0.35-0.90	Delay contribution to deterioration propensity

Note. Reference values and uncertainty ranges are provisional normalized model parameters used for computational exploration and sensitivity analysis, they are not patient-estimated clinical effect sizes.

Delay load was defined piecewise as

$$D(t) = \min\left(\frac{t}{s_D}, 1.5\right), for\ t < \tau,$$

and

$$D(t) = \left(\frac{\tau}{s_D}\right)\exp[-0.7(t-\tau)], for\ t \geq \tau.$$

Thus, delay burden accumulated before intervention and subsequently decayed rather than disappearing instantaneously. The intervention term represents a generic definitive surgical/perioperative correction of the dominant perfusion disturbance and is not intended to represent a particular operation, diagnosis, or treatment protocol. The equations encode hypothesized directionality and coupling rather than clinically estimated causal effects. This distinction is essential because the physiological evidence underlying the observable markers derives largely from pediatric shock and sepsis cohorts rather than a homogeneous emergency-surgical population (Choudhary et al., 2017, Fernandez-Sarmiento et al., 2024 and Weiss et al., 2020).

2.4 Latent Regime-Switching Model

To represent qualitative changes in physiological state, the continuous dynamical system was coupled to a four-state continuous-time Markov chain comprising compensated, vulnerable, decompensating, and refractory regimes. Let

$$p(t) = [p_0(t), p_1(t), p_2(t), p_3(t)]$$

denote the corresponding state-probability vector. Its evolution was governed by the state-dependent Kolmogorov forward equation

$$\frac{dp(t)}{dt} = p(t)Q(t),$$

where $Q(t)$ is the infinitesimal generator matrix (Norris, 1997). A composite deterioration driver was defined as

$$z(t) = 0.34L_n(t) + 0.22C_n(t) + 0.26[1 - P(t)] + 0.28[1 - R(t)] + 0.20I(t) + 0.18d(t),$$

where

$$d(t) = \frac{\min(t, \tau)}{s_D}$$

and s_D denotes the delay-scaling parameter. The state-dependent forward transition intensities were

$$q_{01}(t) = 0.06 \exp[1.7z(t)]$$

$$q_{12}(t) = 0.05 \exp\{2.0[z(t) - 0.22]$$

and

$$q_{23}(t) = 0.035 \exp\{2.3[z(t) - 0.38]$$

whereas reverse transition intensities were defined as

$$q_{10}(t) = 0.035 \exp\{1.5[R(t) - 0.5]$$

$$q_{21}(t) = 0.026 \exp\{1.4[P(t) - 0.5]$$

and

$$q_{32}(t) = 0.018 \exp\{1.3[R(t) - 0.4]$$

Accordingly, the generator matrix was

$$Q(t) = \begin{pmatrix} -q_{01} & q_{01} & 0 & 0 \\ q_{10} & -(q_{10} + q_{12}) & q_{12} & 0 \\ 0 & -q_{21} & -(q_{21} + q_{23}) & q_{23} \\ 0 & 0 & q_{32} & -q_{32} \end{pmatrix}$$

The initial regime-probability vector was

$$p(0) = (0.68, 0.24, 0.07, 0.01)$$

corresponding respectively to the compensated, vulnerable, decompensating, and refractory regimes. All off-diagonal transition intensities are non-negative, and each diagonal element is the negative sum of the remaining elements in its row, ensuring that each row of $Q(t)$ sums to zero and preserving total probability under the continuous-time Markov formulation.

The transition structure was designed so that increasing normalized lactate burden, prolonged capillary refill, declining perfusion and recovery reserves, increasing injury burden, and accumulated delay increase the modelled tendency toward more compromised regimes. Conversely, preserved perfusion and recovery reserves increase reverse-transition intensities toward less compromised regimes. These transition functions and coefficients are provisional structural assumptions of the proof-of-concept model rather than transition rates estimated from patient-level data. The four Markov states therefore represent latent computational regimes and should not be interpreted as validated bedside classifications.

2.5 Deterioration Propensity and Risk Propagation Layer

A bounded logistic score summarized the instantaneous modeled propensity toward deterioration: $H(t) = logit^{-1}[-3 + \beta_L L_n + \beta_C C_n + \beta_R(1-R) + \beta_D D]$. This quantity was used as an internal trajectory-comparison score rather than a formal survival hazard or calibrated event probability. The direction of the delay contribution was motivated by evidence that operative delay is associated with adverse outcomes in pediatric emergency surgery (Butterworth et al., 2023 and Kaushal-Deep et al., 2019). However, β_D was not fitted to either cohort and does not reproduce a published odds ratio, hazard ratio, or absolute clinical risk. This terminology is deliberately restricted to deterioration propensity because the present proof-of-concept does not specify an observed event-time likelihood or estimate a clinical hazard function.

2.6 Dynamic Recovery Index and Reversibility Classes

A multiplicative recovery index was defined as

$$G(t) = \left[P(t)R(t)\left(1 - I(t)\right)\right]^{\frac{1}{3}},$$

which assigns equal multiplicative contribution to perfusion reserve, recovery reserve, and absence of accumulated injury. For proof-of-concept classification, terminal states satisfying $P \geq 0.75, R \geq 0.60$, and $I \leq 0.25$ were classified as reversible. States satisfying $P \geq 0.58, R \geq 0.45$, and $I \leq 0.42$ without meeting the reversible criteria were classified as partially reversible. All remaining terminal states were classified as irreversible. These boundaries are structural simulation assumptions and are not clinical thresholds. The reversible-window boundary was defined as $\tau_{RW} = \sup\{\tau : x(T;\tau)\ belongs\ to\ the\ reversible\ class\}$, where T denotes the specified simulation horizon. The Dynamic Recovery Window therefore represents a model-derived property of the assumed equations and parameters rather than a recommended safe waiting time.

2.7 Optimal Stopping Decision Layer

A separate finite-horizon decision layer was used to examine intervention timing mathematically. For each candidate intervention delay \(\tau\), the policy value was defined as

$$V(\tau) = 0.40P_T + 0.34R_T + 0.18(1 - I_T) - 0.08\log(1 + L_T) - c_w\tau - c_I[1 + 0.6I(\tau)]$$

where P_T, R_T, I_T, and L_T denote perfusion reserve, recovery reserve, injury burden, and lactate at the 10-h simulation horizon, respectively. The waiting-cost coefficient was $c_w = 0.045$, and the intervention-cost coefficient was $c_I = 0.08$. The injury-dependent term $I(\tau)$ allowed the intervention penalty to vary with the modeled injury burden at the candidate intervention time. These weights and penalties were provisional decision-analytic assumptions and were not estimated from patient outcomes. Candidate intervention delays were evaluated over the discrete grid

$$\tau \in \{0, 0.5, 1.0, \ldots, 4.0\}\ h,$$

and the modeled optimal stopping time was defined as

$$\tau_{OS}^{*} = \arg\max_{\tau} V(\tau)$$

Intervention timing was therefore treated as a finite-horizon discrete optimal-stopping problem in which alternative intervention times were compared through a state-dependent objective function (Peskir & Shiryaev, 2006). The resulting τ_{OS}^{*} represents an internal property of the assumed dynamical and utility structure rather than a clinically recommended intervention time.

The Dynamic Recovery Window boundary τ_{RW} and the optimal-stopping solution τ_{OS}^{*}represent distinct mathematical quantities. The former identifies the latest evaluated delay remaining within the predefined reversible class, whereas the latter identifies the candidate delay that maximizes the provisional decision objective. The two quantities were therefore evaluated and interpreted separately. The decision layer draws conceptually on reserve depletion, regime switching, uncertainty, and irreversible stopping decisions, these concepts are used as mathematical analogies and do not imply equivalence between financial and physiological systems.

2.8 Mathematical and Structural Analysis

For finite parameter values, the continuous vector field is locally continuously differentiable within the interior of the modeled domain, supporting local existence and uniqueness of solutions between intervention-induced changes in $u(t)$. Because $u(t)$ and $D(t)$ change at the intervention time, trajectories were treated piecewise across the pre- and post-intervention intervals with continuity of the physiological state variables at $t = \tau$. The continuous equations were examined at relevant state boundaries to assess directional consistency. At $P = 0$ and $R = 0$, restoration terms are non-negative; at $P = 1$ and $R = 1$, restoration terms vanish while depletion terms oppose further outward growth. The injury source contains the factor $(1 - I)$, whereas its clearance term is non-negative for admissible states.

In the numerical implementation, P, R, and I were additionally constrained to $[0,1]$. This computational projection was treated as a numerical safeguard and was distinguished from analytical claims of positive invariance. Delay-related parameters, reperfusion gain, lactate clearance, reserve restoration, and injury-clearance rates were treated as continuation or perturbation parameters because

changes in these quantities can alter local behavior and terminal recovery classification. Full structural or practical identifiability is not claimed. In future observed-data applications, lactate and capillary refill would constitute measured variables, whereas P, R, and I would require explicit measurement or state-space observation models.

2.9 Computational Analysis and Robustness

Reference trajectories were simulated over a 10-hour horizon using fixed-step fourth-order Runge-Kutta integration with $\Delta t = 0.2h$. Numerical robustness was assessed using adaptive RK45 and DOP853 methods and the implicit Radau and BDF solvers. Delay continuation was evaluated for intervention times from 0 to 6 hours in 0.5-hour increments. Seven clinically interpretable scenarios altered lactate clearance, reperfusion gain, post-intervention injury clearance, baseline lactate, baseline capillary refill, or initial reserve while retaining the remaining reference assumptions.

Global parameter uncertainty was evaluated using Latin-hypercube samples across the 20 predefined parameter ranges in Table 2, providing stratified coverage of the multidimensional parameter space (McKay et al., 1979). Partial rank correlation coefficients were then calculated for the terminal recovery index at a 2-hour intervention delay after rank-based adjustment for the remaining sampled parameters (Marino et al., 2008).

A 15×15 grid spanning initial lactate and recovery reserve was evaluated for the optimal-stopping analysis. For each grid point, the candidate delay maximizing the provisional decision objective was recorded together with its corresponding policy value. Under the current parameterization, the resulting stopping decision must be interpreted jointly with the actual computational output rather than assumed to contain a non-zero waiting region. Parameter influence on the terminal recovery index was quantified using partial rank correlation coefficients within the global uncertainty framework described for systems biology models by Marino et al. (2008).

Finally, an internal synthetic model-recovery experiment comprising simulated states compared five nested logistic representations using Akaike's information criterion (AIC). Nested synthetic representations were compared using Akaike's information criterion (AIC), which balances model fit against the number of estimated parameters (Akaike, 1974). The experiment was designed solely to determine whether information from the generating delay-sensitive structure could be recovered after deliberate model reduction. Because the outcomes were generated from the proposed structural model, differences in AIC or synthetic classification accuracy were not interpreted as evidence of clinical predictive superiority.

3. Results

3.1 Structural and Numerical Verification

The simulation remained finite throughout the 10-hour horizon, with the normalized latent states remaining within their prescribed computational domains. The four latent regime probabilities remained non-negative and summed to one to numerical precision, consistent with the generator-

matrix construction of the continuous-time Markov chain. The terminal recovery index obtained with fixed-step fourth-order Runge-Kutta integration was 0.9603, compared with approximately 0.9599 using RK45, DOP853, Radau, and BDF. Terminal perfusion reserve differed by less than 0.0001 across the adaptive solvers. These differences were negligible relative to the scale of the simulated state variables, indicating that the principal reference trajectory and terminal recovery classification were numerically robust to the choice of integrator.

3.2 Reference Trajectory

Under the reference scenario with intervention at 1 hour, lactate decreased from 4.20 mmol/L at baseline to 1.52 mmol/L at 10 hours, corresponding to a modeled reduction of 63.9%. Capillary refill time decreased from 3.10 to 1.52 seconds. Perfusion reserve initially declined from 0.580 to a minimum of 0.573 before increasing to 0.998, whereas recovery reserve declined from 0.620 to a minimum of 0.589 before recovering to 0.915. Injury burden increased transiently to a peak of 0.396 and subsequently declined to 0.030. The deterioration-propensity score reached a maximum of 0.111 and decreased to 0.061 by the end of the simulation. The transient decline in perfusion and recovery reserve together with the early increase in injury burden illustrates that intervention did not instantaneously reset the modeled physiological state; recovery emerged progressively from the coupled post-intervention dynamics. These values represent properties of the reference equations and provisional parameterization rather than patient-level treatment effects.

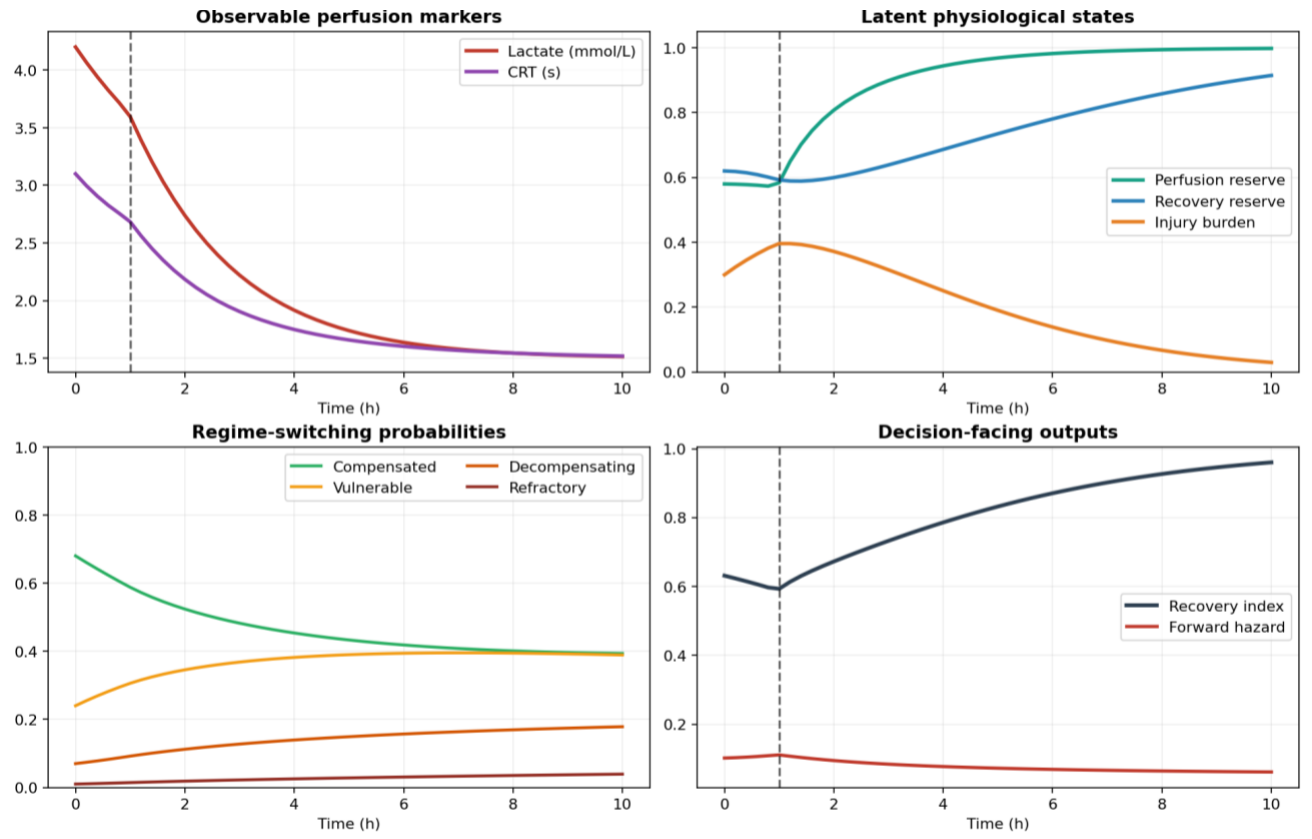


Figure 1. *Reference theoretical trajectory under intervention at 1 hour. The vertical dashed line indicates the modeled intervention time. Observable and latent trajectories are simulation outputs under the reference parameterization.*

3.3 Delay Continuation and Dynamic Recovery Window

Delay continuation from 0 to 6 hours in 0.5-hour increments produced three recovery classes under the reference parameterization. Intervention delays from 0 to 4.5 hours terminated within the predefined reversible basin, delays of 5.0 and 5.5 hours terminated within the partially reversible basin, and a 6.0-hour delay entered the irreversible class. The resulting reference Dynamic Recovery Window was therefore 4.5 hours under the stated equations, parameter values, classification criteria, and simulation horizon. The terminal recovery index declined monotonically as intervention was delayed. At 6 hours after presentation, intervention at 1 hour produced modeled lactate of 1.64 mmol/L and capillary refill of 1.60 seconds, whereas intervention delayed to 4 hours resulted in corresponding values of 2.05 mmol/L and 1.96 seconds. When intervention was delayed to 6 hours, lactate and capillary refill remained at 2.71 mmol/L and 2.54 seconds, respectively, at the 6-hour time point. These trajectories show that the modeled recovery classification emerged from accumulated changes in perfusion reserve, recovery reserve, injury burden, and observable perfusion markers rather than from a binary indicator of elapsed delay. The 4.5-hour boundary is therefore a property of the reference simulation and should not be interpreted as a clinically safe waiting time.

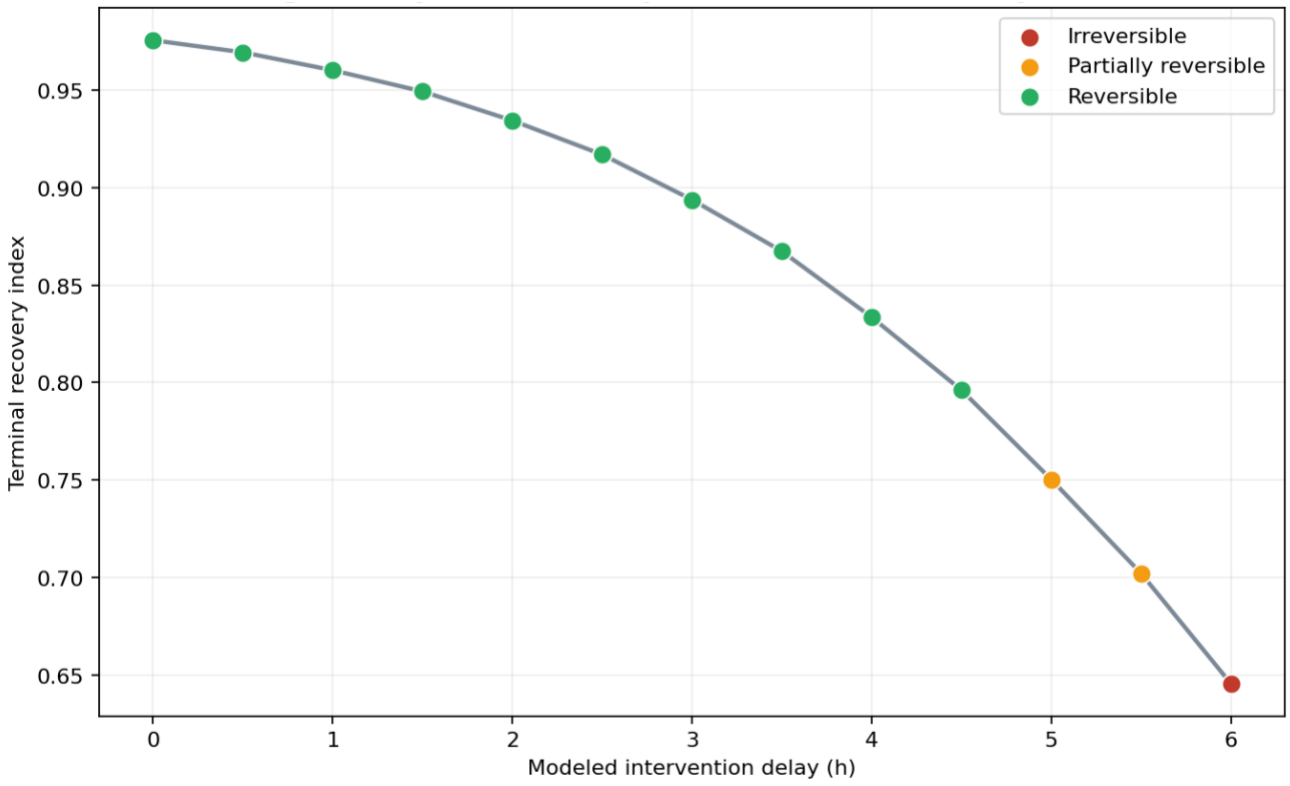


Figure 2*. Delay continuation analysis and dynamic recovery window under the reference parameterization. Simulated trajectories remained within the predefined reversible class through 4.5 hours, entered the partially reversible class at 5.0-5.5 hours, and entered the irreversible class at 6.0 hours. These boundaries are model classifications, not clinical cut-offs.*

3.4 Scenario Stress Testing

Table 3. Scenario-based simulation results under predefined parameter perturbations.

Scenario	Reversible window (h)	Partial window (h)	Terminal recovery index	Terminal perfusion	Terminal reserve	Terminal injury	Peak deterioration propensity	Lactate at 6 h	CRT at 6 h
Reference	4.500	5.500	0.960	0.998	0.915	0.030	0.111	1.638	1.604
Faster lactate clearance	4.500	5.500	0.963	0.999	0.918	0.026	0.111	1.571	1.599
Stronger reperfusion	4.500	5.500	0.963	0.998	0.919	0.027	0.111	1.630	1.595
Slower injury resolution	3.500	5.000	0.932	0.997	0.876	0.074	0.111	1.639	1.624
Higher initial lactate	4.000	5.500	0.951	0.997	0.898	0.039	0.132	1.707	1.626
Prolonged capillary refill	4.000	5.500	0.951	0.997	0.898	0.039	0.130	1.659	1.635
Low reserve profile	4.000	5.000	0.942	0.997	0.881	0.048	0.139	1.675	1.646

Slower post-intervention injury resolution produced the largest contraction of the reversible window, reducing it from 4.5 to 3.5 hours and lowering the terminal recovery index from 0.960 to 0.932. Higher initial lactate, prolonged capillary refill, and the low-reserve profile each reduced the reversible window to 4.0 hours. The low-reserve profile also produced the highest peak deterioration-propensity score (0.139), followed by higher initial lactate (0.132) and prolonged capillary refill (0.130). In contrast, faster lactate clearance and stronger reperfusion increased the terminal recovery index to 0.963 but did not extend the reversible boundary beyond 4.5 hours on the 0.5-hour delay grid. Within the tested scenarios, recovery classification was therefore more sensitive to injury-resolution and reserve-related perturbations than to isolated improvement of a single observable perfusion marker.

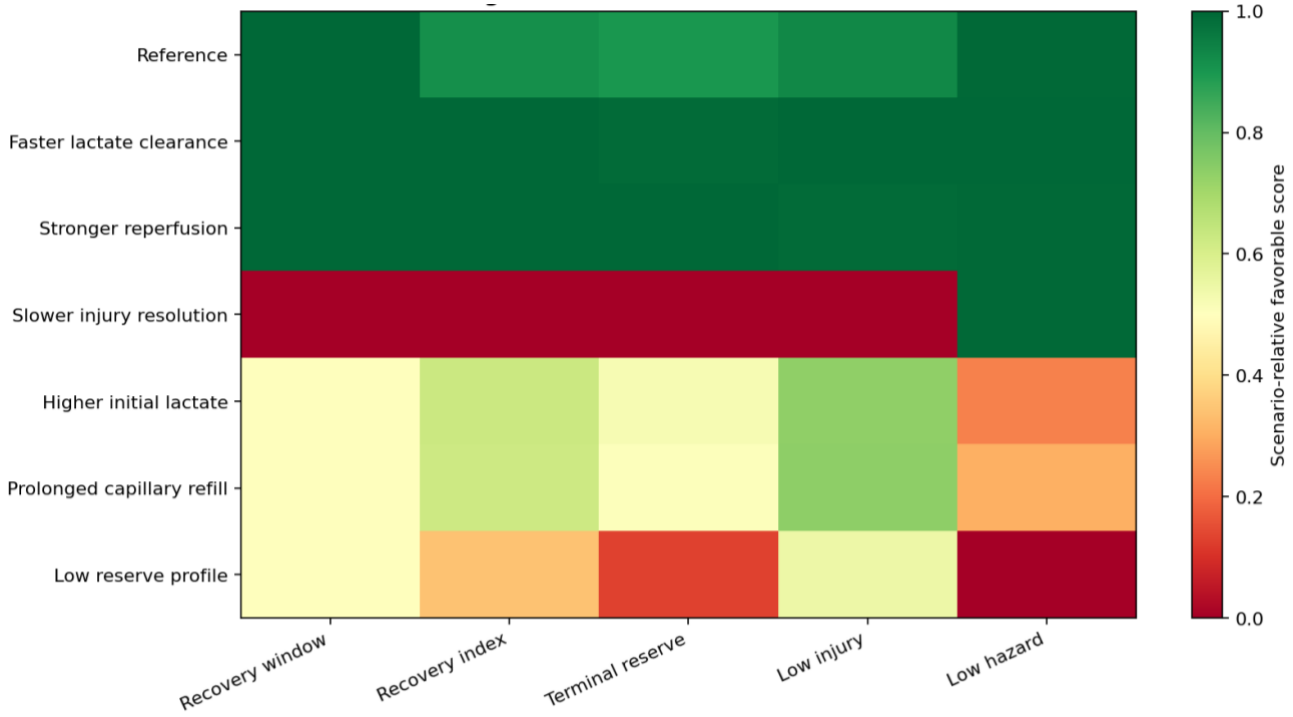


Figure 3. *Scenario stress-test matrix across the predefined perturbation scenarios. Displayed scores are normalized within the simulated scenarios for visualization and do not represent patient probabilities.*

3.5 Global Sensitivity and Uncertainty

Across Latin-hypercube parameter sets evaluated at a 2-hour intervention delay, the terminal recovery index had a mean of 0.925, standard deviation of 0.035, median of 0.932, and 2.5^{th} - 97.5^{th} percentile range of 0.840-0.975. The distribution therefore remained concentrated within a relatively high recovery-index range under the predefined parameter uncertainty, although substantial variation remained across sampled parameter combinations. Partial rank correlation analysis identified recovery-reserve restoration (PRCC = 0.933) and post-intervention injury clearance (PRCC = 0.923) as showed the strongest positive associations with terminal recovery. Reserve injury (PRCC = -0.833) and injury generation (PRCC = -0.775) were the strongest adverse determinants. Reperfusion gain (PRCC = 0.505), lactate-associated perfusion penalty (PRCC = -0.356), reserve depletion associated with delay (PRCC = -0.343), post-intervention lactate clearance (PRCC = 0.314), baseline perfusion restoration (PRCC = 0.299), and capillary-refill recovery (PRCC = 0.297) formed a second tier of sensitivity. These results indicate that, within the sampled parameter ranges, the terminal recovery index was most sensitive to processes governing reserve restoration and injury resolution rather than to variation in a single observable perfusion marker. The direction of the dominant PRCCs was consistent with the model structure: processes restoring reserve or clearing injury increased recovery, whereas processes generating injury or depleting reserve reduced it.

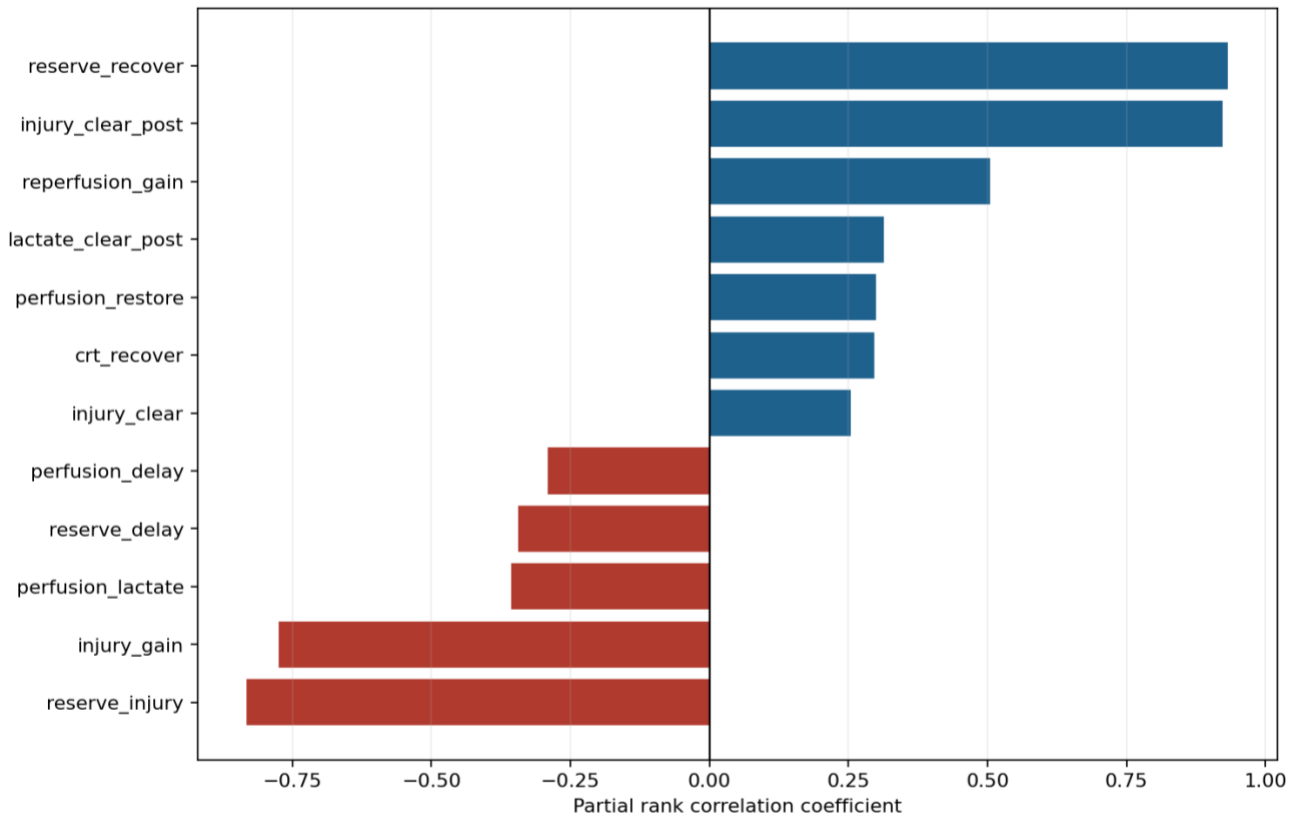


Figure 4. *Global sensitivity of the terminal recovery index at a 2-hour intervention delay. Bars show partial rank correlation coefficients derived from 1,000 Latin-hypercube parameter sets. Positive values indicate association with greater terminal recovery within the simulated parameter space, whereas negative values indicate association with lower terminal recovery.*

3.6 Optimal Stopping Analysis

Across the 15×15 grid of initial lactate and recovery-reserve states, the provisional decision objective selected immediate intervention ($\tau_{OS} * = 0\ h$) for all 225 evaluated combinations. Thus, under the current utility weights, waiting penalty, intervention cost, and reference dynamical parameters, the optimal-stopping analysis did not produce a non-zero waiting region. Although the selected delay was invariant across the evaluated grid, the corresponding policy value varied with initial lactate and recovery reserve, indicating that initial physiological state affected the value assigned to the intervention policy even when it did not alter the maximizing intervention time. The absence of a non-zero waiting region suggests that, under the current decision specification, the cost assigned to continued delay dominates any modeled value of waiting. This finding should be interpreted as a diagnostic property of the provisional decision layer rather than evidence that immediate surgery is clinically optimal. The stopping objective was not calibrated against patient outcomes, and its utility weights and penalties remain structural assumptions. The result therefore identifies the decision layer as a component requiring empirical calibration before a clinically meaningful state-dependent stopping boundary can be evaluated.

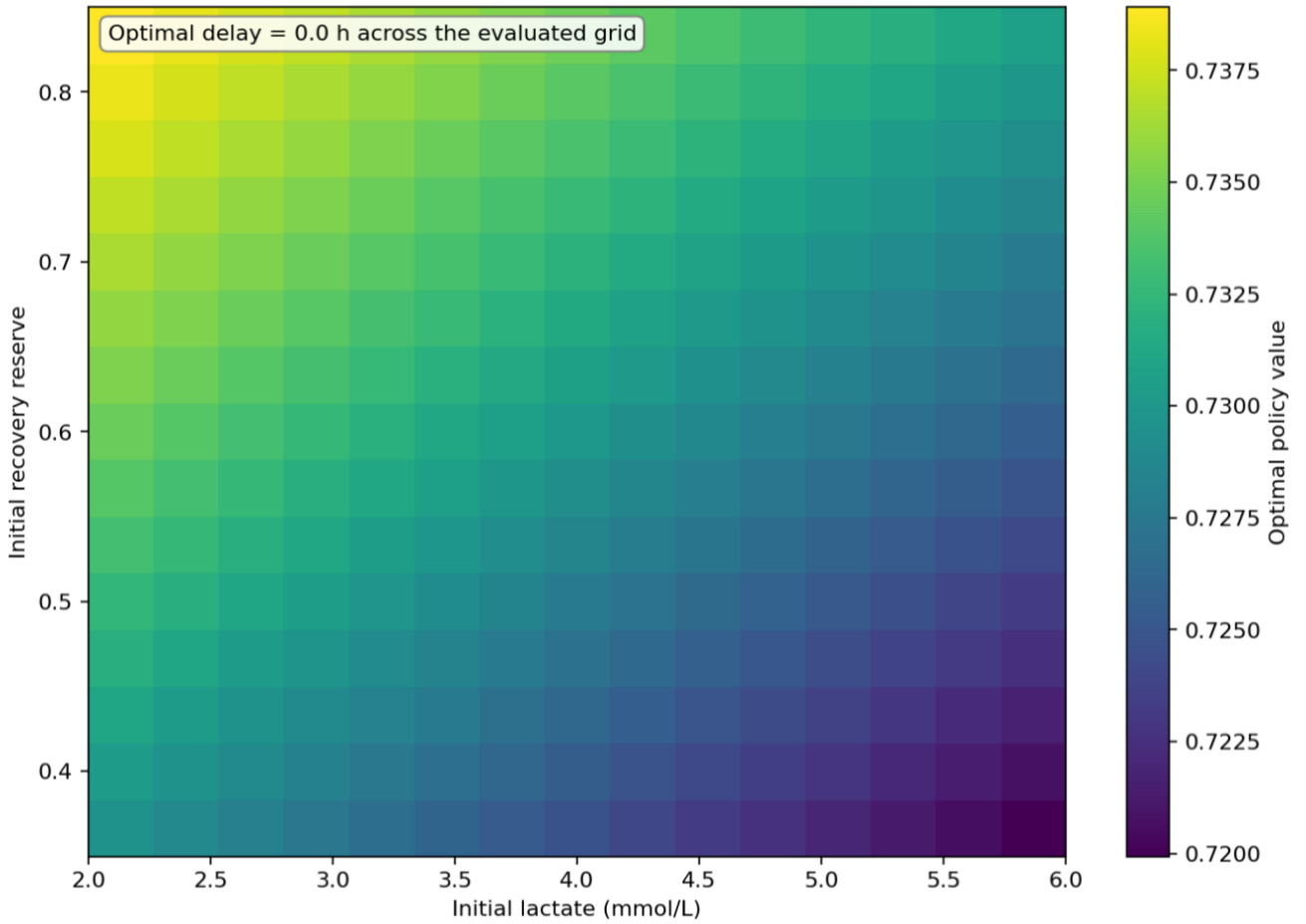


Figure 5. *Optimal-stopping policy surface across initial lactate and recovery reserve. Under the current provisional objective, the maximizing intervention delay was 0 h across all 225 evaluated states; color variation represents the corresponding policy value rather than variation in optimal delay.*

3.7 Internal Synthetic Model-Recovery Analysis

The full delay-sensitive representation produced the lowest AIC (5468.2) and the highest synthetic classification accuracy (0.720), followed by the clinical five-variable representation (AIC 5819.8; accuracy 0.700) and the perfusion-reserve representation (AIC 5836.1; accuracy 0.697). The lactate-only representation produced the highest AIC (6371.5) and lowest synthetic accuracy (0.635).

Table 4. Internal Synthetic AIC model recovery comparison.

Model	Variables	AIC	Synthetic Accuracy
Full delay-sensitive state	6	5468.2	0.720
Clinical five variable	5	5819.8	0.700
Perfusion reserve model	4	5836.1	0.697
Lactate plus CRT	2	5993.3	0.679
Lactate only	1	6371.5	0.635

This ordering is expected because the synthetic outcomes were generated from the full structural rule. The analysis therefore evaluates recovery of the generating information structure and quantifies

information loss after deliberate model reduction. It does not show superiority over established clinical prediction models or provide evidence of patient-level predictive performance.

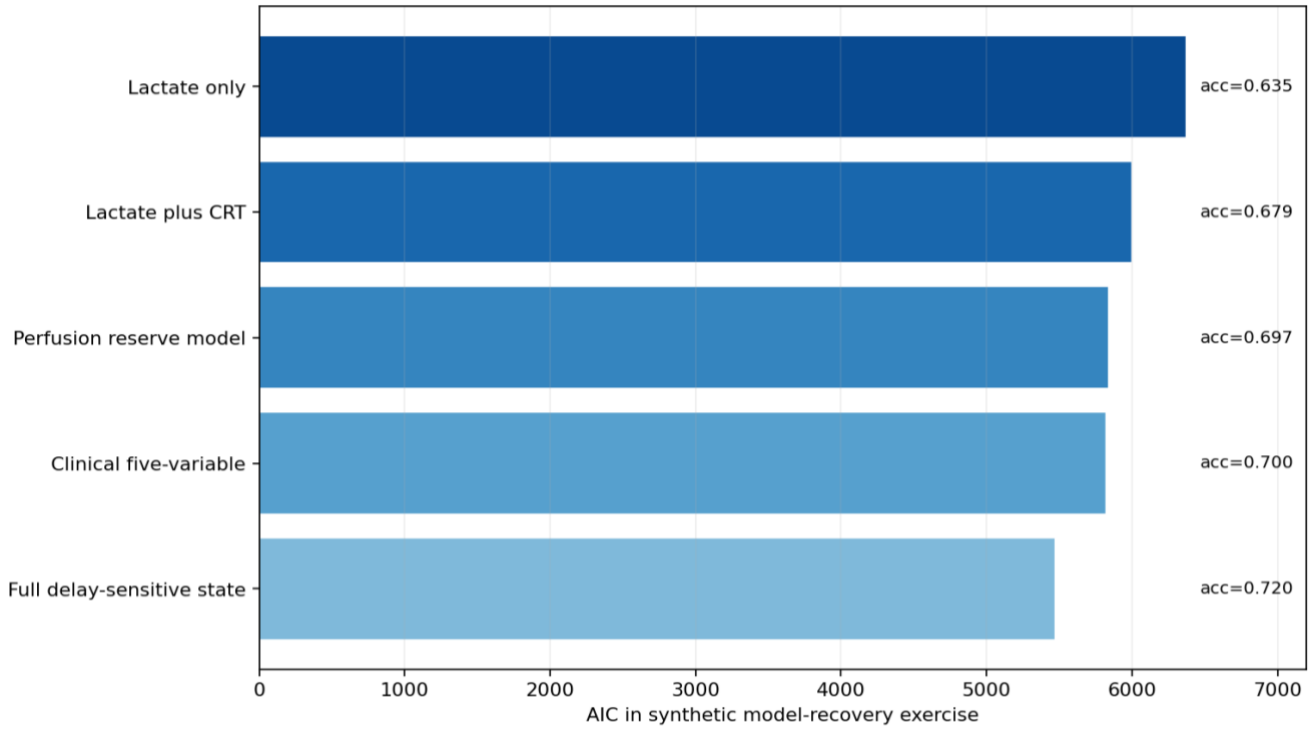


***Figure 6**. Internal synthetic model recovery comparison using Akaike's information criterion. Accuracy labels refer exclusively to the synthetic generating experiment and should not be interpreted as clinical predictive performance.*

4. Discussion

This study shows that intervention delay in pediatric emergency surgery can be represented as a dynamic recovery problem rather than as time alone. Under the reference parameterization, the simulated trajectory remained within the predefined reversible class through 4.5 hours, became partially reversible at 5.0-5.5 hours, and entered the irreversible class at 6.0 hours. These boundaries arose from the joint evolution of lactate, capillary refill, perfusion reserve, recovery reserve, and injury burden and should be interpreted as model outputs rather than clinical thresholds. The scenario and sensitivity analyses showed that recovery depended more strongly on reserve restoration and injury resolution than on isolated improvement in an observable marker. Slower injury resolution shortened the reversible window from 4.5 to 3.5 hours, whereas higher initial lactate, prolonged capillary refill, and low reserve reduced it to 4.0 hours. Consistently, recovery-reserve restoration (PRCC = 0.933) and post-intervention injury clearance (PRCC = 0.923) were the strongest favorable determinants, while reserve injury (PRCC = -0.833) and injury generation (PRCC = -0.775) were the strongest adverse determinants. These associations describe sensitivity within the simulated parameter space and are not clinical effect estimates.

The dynamic recovery window and optimal-stopping analysis provided complementary but distinct information. The recovery-window analysis identified the latest simulated delay remaining within the predefined reversible basin, whereas the provisional optimal stopping objective selected immediate intervention ($\tau * = 0\ h$) across all 225 lactate recovery reserve combinations. This does not establish immediate intervention as clinically optimal; rather, it indicates that continued waiting was not favored under the current provisional utility weights and penalties. Clinical calibration would be required before interpreting the stopping layer as a timing rule. The computational checks supported the internal robustness of the framework. Terminal recovery indices were nearly identical across RK4, RK45, DOP853, Radau, and BDF solvers, while the synthetic AIC analysis recovered the expected advantage of the generating delay-sensitive structure. These findings support numerical and computational consistency but do not constitute biological or clinical validation.

Overall, the model provides a mechanistic representation connecting observable perfusion markers with latent reserve, injury accumulation, regime switching, and intervention timing. Its main contribution is therefore not a specific safe-delay threshold, but a testable representation of how recoverability may evolve during waiting. Future patient-level studies should calibrate the latent states, transition parameters, recovery boundaries, and decision weights before the framework is considered for clinical prediction or decision support.

5. Limitations

This study has several limitations. First, no patient level calibration was performed; therefore, model coefficients, latent state parameters, and reversibility boundaries remain provisional. Second, much of the physiological evidence was derived from pediatric shock and sepsis studies rather than a homogeneous emergency surgical population, requiring direct calibration in pediatric surgical cohorts. Third, capillary refill is sensitive to measurement technique and environmental conditions and will require standardized assessment in future studies. Fourth, the intervention term represents a generic physiological correction and does not distinguish among specific surgical conditions or procedures. Fifth, perfusion reserve, recovery reserve, and injury burden are latent constructs that currently lack patient level observation models and will require repeated clinical measurements for estimation. Sixth, the optimal stopping objective was not clinically calibrated and selected immediate intervention across the evaluated grid under the current provisional utility weights and penalties. This result should not be interpreted as a clinical timing recommendation. Finally, the synthetic AIC analysis evaluates recovery of the model's own generating structure and does not provide external predictive validation. Consequently, the simulated recovery window and stopping policy should be regarded as mathematical proof-of-concept outputs rather than clinical safe-delay thresholds.

6. Conclusion

The dynamic recovery window model provides a mathematical proof-of-concept for representing intervention delay in pediatric surgical emergencies as a time dependent interaction among perfusion markers, recovery reserve, injury accumulation, latent state transitions, and intervention timing. The simulations showed that recoverability changed progressively with delay and was particularly sensitive to reserve restoration and injury resolution, while the provisional optimal

stopping objective favored immediate intervention across the evaluated state space. These findings show the potential of coupled dynamical and decision models for studying time sensitive recovery, but the resulting recovery window and stopping policy are model derived rather than clinical thresholds. Patient level calibration and external validation are required before the framework can support clinical prediction or surgical decision making.

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